

Introduction to Machine Learning Problems: Logistic Regression

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1. Suggest possible response variables and predictors for the following classification problems. For each problem, indicate how many classes there are. There is no single correct answer.

- (a) Given an audio sample, to detect the gender of the voice.

Solution:

- Response variable: Gender (Male/Female) - 2 classes
- Predictors:
 - Average pitch (fundamental frequency)
 - Voice duration
 - Speaking rate (words per minute)

- (b) A electronic writing pad records motion of a stylus and it is desired to determine which letter or number was written. Assume a segmentation algorithm is already run which indicates very reliably the beginning and end time of the writing of each character.

Solution:

- Response variable: Character class (A-Z, 0-9) - 36 classes
- Predictors:
 - Number of strokes
 - Total writing time
 - Average stroke length

2. Suppose that a logistic regression model for a binary class label $y = 0, 1$ is given by

$$P(y = 1|\mathbf{x}) = \frac{1}{1 + e^{-z}}, \quad z = \beta_0 + \beta_1 x_1 + \beta_2 x_2,$$

where $\beta = [1, 2, 3]^T$. Describe the following sets:

- (a) The set of $\mathbf{x}$ such that $P(y = 1|\mathbf{x}) > P(y = 0|\mathbf{x})$.

Solution: Since $P(y = 0|\mathbf{x}) = 1 - P(y = 1|\mathbf{x})$, we have:

$$P(y = 1|\mathbf{x}) > P(y = 0|\mathbf{x}) \Rightarrow P(y = 1|\mathbf{x}) > 0.5$$

This occurs when $z > 0$, i.e., when:

$$1 + 2x_1 + 3x_2 > 0$$

So the set is: $\{(x_1, x_2) : 2x_1 + 3x_2 > -1\}$

- (b) The set of $\mathbf{x}$ such that $P(y = 1|\mathbf{x}) > 0.8$.

Solution: We need $P(y = 1|\mathbf{x}) > 0.8$, which means:

$$\frac{1}{1 + e^{-z}} > 0.8 \Rightarrow 1 > 0.8(1 + e^{-z}) \Rightarrow 0.2 > 0.8e^{-z} \Rightarrow e^{-z} < 0.25$$

Taking natural log: $-z < \ln(0.25) \Rightarrow z > -\ln(0.25) = \ln(4)$

So: $1 + 2x_1 + 3x_2 > \ln(4) \approx 1.386$

The set is: $\{(x_1, x_2) : 2x_1 + 3x_2 > 0.386\}$

- (c) The set of x_1 such that $P(y = 1|\mathbf{x}) > 0.8$ and $x_2 = 0.5$.

Solution: Substituting $x_2 = 0.5$ into the condition from part (b):

$$2x_1 + 3(0.5) > 0.386 \Rightarrow 2x_1 + 1.5 > 0.386 \Rightarrow 2x_1 > -1.114 \Rightarrow x_1 > -0.557$$

The set is: $\{x_1 : x_1 > -0.557\}$

3. A data scientist is hired by a political candidate to predict who will donate money. The data scientist decides to use two predictors for each possible donor:

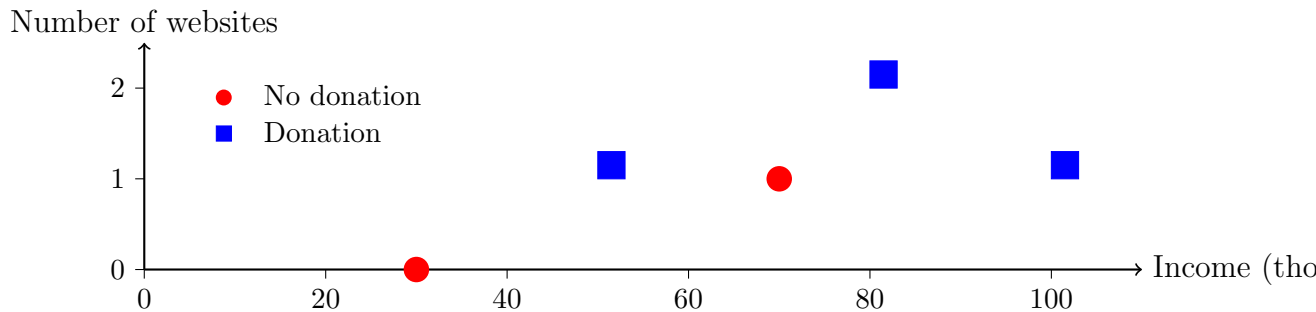
- x_1 = the income of the person (in thousands of dollars), and
- x_2 = the number of websites with similar political views as the candidate the person follow on Facebook.

To train the model, the scientist tries to solicit donations from a randomly selected subset of people and records who donates or not. She obtains the following data:

Income (thousands \$), x_{i1}	30	50	70	80	100
Num websites, x_{i2}	0	1	1	2	1
Donate (1=yes or 0=no), y_i	0	1	0	1	1

- (a) Draw a scatter plot of the data labeling the two classes with different markers.

Solution:



- (b) Find a linear classifier that makes at most one error on the training data. The classifier should be of the form,

$$\hat{y}_i = \begin{cases} 1 & \text{if } z_i > 0 \\ 0 & \text{if } z_i < 0, \end{cases} \quad z_i = \mathbf{w}^\top \mathbf{x}_i + b.$$

What is the weight vector $\mathbf{w}$ and bias b in your classifier?

Solution: Looking at the data, I will use a classifier that incorporates both features:

$$z_i = 2x_{i1} + 20x_{i2} - 110$$

This gives us $\mathbf{w} = [2, 20]^\top$ and $b = -110$.

Checking:

- Sample 1: $z_1 = 2(30) + 20(0) - 110 = -50 < 0 \Rightarrow \hat{y}_1 = 0$ (correct)
- Sample 2: $z_2 = 2(50) + 20(1) - 110 = 10 > 0 \Rightarrow \hat{y}_2 = 1$ (correct)
- Sample 3: $z_3 = 2(70) + 20(1) - 110 = 50 > 0 \Rightarrow \hat{y}_3 = 1$ (error)
- Sample 4: $z_4 = 2(80) + 20(2) - 110 = 90 > 0 \Rightarrow \hat{y}_4 = 1$ (correct)
- Sample 5: $z_5 = 2(100) + 20(1) - 110 = 110 > 0 \Rightarrow \hat{y}_5 = 1$ (correct)

This makes only 1 error on sample 3.

(c) Now consider a logistic model of the form,

$$P(y_i = 1|\mathbf{x}_i) = \frac{1}{1 + e^{-z_i}}, \quad z_i = \mathbf{w}^\top \mathbf{x}_i + b.$$

Using $\mathbf{w}$ and b from the previous part, which sample i is the *least* likely (i.e. $P(y_i|\mathbf{x}_i)$ is the smallest). If you do the calculations correctly, you should not need a calculator.

Solution: Using $\mathbf{w} = [2, 20]^\top$ and $b = -110$, we have $z_i = 2x_{i1} + 20x_{i2} - 110$.

For each sample, we need to compute $P(y_i|\mathbf{x}_i)$, which is $P(y_i = 1|\mathbf{x}_i)$ if $y_i = 1$ and $P(y_i = 0|\mathbf{x}_i)$ if $y_i = 0$.

Computing z_i values:

- Sample 1 ($y_1 = 0$): $z_1 = -50$, so $P(y_1 = 0|\mathbf{x}_1) = 1 - \frac{1}{1+e^{50}} \approx 1$ (very likely)
- Sample 2 ($y_2 = 1$): $z_2 = 10$, so $P(y_2 = 1|\mathbf{x}_2) = \frac{1}{1+e^{-10}} \approx 1$ (very likely)
- Sample 3 ($y_3 = 0$): $z_3 = 50$, so $P(y_3 = 0|\mathbf{x}_3) = 1 - \frac{1}{1+e^{-50}} \approx 0$ (very unlikely)
- Sample 4 ($y_4 = 1$): $z_4 = 90$, so $P(y_4 = 1|\mathbf{x}_4) = \frac{1}{1+e^{-90}} \approx 1$ (very likely)
- Sample 5 ($y_5 = 1$): $z_5 = 110$, so $P(y_5 = 1|\mathbf{x}_5) = \frac{1}{1+e^{-110}} \approx 1$ (very likely)

Sample 3 is the least likely because the model strongly predicts it should donate ($z_3 = 50$ is large and positive), but it actually did not donate ($y_3 = 0$). So $P(y_3 = 0|\mathbf{x}_3) \approx 0$.

(d) Now consider a new set of parameters

$$\mathbf{w}' = \alpha \mathbf{w}, \quad b' = \alpha b,$$

where $\alpha > 0$ is a positive scalar. Would using the new parameters change the values $\hat{y}$ in part (b)? Would they change the likelihoods $P(y_i|\mathbf{x}_i)$ in part (c)? If they do not change, state why. If they do change, qualitatively describe the change as a function of α .

Solution:

- The $\hat{y}$ values would NOT change because the decision boundary $z_i = 0$ remains the same: $\alpha z_i = 0$ has the same solution as $z_i = 0$ (since $\alpha > 0$).

- The likelihoods $P(y_i|\mathbf{x}_i)$ WOULD change. Since $z'_i = \alpha z_i$, we have:

$$P(y_i = 1|\mathbf{x}_i) = \frac{1}{1 + e^{-\alpha z_i}}$$

As α increases:

- If $z_i > 0$ (originally high probability), $P(y_i = 1|\mathbf{x}_i)$ approaches 1 more quickly
- If $z_i < 0$ (originally low probability), $P(y_i = 1|\mathbf{x}_i)$ approaches 0 more quickly

So larger α makes the probabilities more extreme (closer to 0 or 1), increasing confidence in predictions.

4. Suppose we collect data for a group of students in a machine learning class with variables X_1 = hours studied, X_2 = undergrad GPA, and Y = receive an A. We fit a logistic regression and produce estimated coefficient, $\beta_0 = -6$, $\beta_1 = 0.05$, $\beta_2 = 1$.

- (a) Estimate the probability that a student who studies for 40 h and has an undergrad GPA of 3.5 gets an A in the class.

Solution: The logistic model is:

$$P(Y = 1|X_1, X_2) = \frac{1}{1 + e^{-z}}, \quad z = \beta_0 + \beta_1 X_1 + \beta_2 X_2$$

For $X_1 = 40$ and $X_2 = 3.5$:

$$z = -6 + 0.05(40) + 1(3.5) = -6 + 2 + 3.5 = -0.5$$

$$P(Y = 1|40, 3.5) = \frac{1}{1 + e^{0.5}} = \frac{1}{1 + 1.649} = \frac{1}{2.649} \approx 0.378$$

- (b) How many hours would the student in part (a) need to study to have a 50 % chance of getting an A in the class?

Solution: For 50% chance, we need $P(Y = 1|X_1, 3.5) = 0.5$, which means $z = 0$.

$$0 = -6 + 0.05X_1 + 1(3.5)$$

$$0 = -6 + 0.05X_1 + 3.5$$

$$0 = -2.5 + 0.05X_1$$

$$0.05X_1 = 2.5$$

$$X_1 = \frac{2.5}{0.05} = 50$$

So the student needs to study 50 hours.

5. The loss function for logistic regression for binary classification is the binary cross entropy defined as

$$J(\beta) = \sum_{i=1}^N \ln(1 + e^{z_i}) - y_i z_i$$

where $z_i = \beta_0 + \beta_1 x_{1,i} + \beta_2 x_{2,i}$ for two features $x_{1,i}$ and $x_{2,i}$.

- (a) What are the partial derivatives of z_i with respect to β_0 , β_1 , and β_2 .

Solution: Since $z_i = \beta_0 + \beta_1 x_{1i} + \beta_2 x_{2i}$:

$$\begin{aligned}\frac{\partial z_i}{\partial \beta_0} &= 1 \\ \frac{\partial z_i}{\partial \beta_1} &= x_{1i} \\ \frac{\partial z_i}{\partial \beta_2} &= x_{2i}\end{aligned}$$

- (b) Compute the partial derivatives of $J(\boldsymbol{\beta})$ with respect to β_0 , β_1 , and β_2 . You should use the chain rule of differentiation.

Solution: Using the chain rule:

$$\frac{\partial J}{\partial \beta_j} = \sum_{i=1}^N \frac{\partial J}{\partial z_i} \frac{\partial z_i}{\partial \beta_j}$$

First, compute $\frac{\partial J}{\partial z_i}$:

$$\frac{\partial J}{\partial z_i} = \frac{\partial}{\partial z_i} [\ln(1 + e^{z_i}) - y_i z_i] = \frac{e^{z_i}}{1 + e^{z_i}} - y_i = \frac{1}{1 + e^{-z_i}} - y_i$$

Now apply the chain rule:

$$\begin{aligned}\frac{\partial J}{\partial \beta_0} &= \sum_{i=1}^N \left(\frac{1}{1 + e^{-z_i}} - y_i \right) \\ \frac{\partial J}{\partial \beta_1} &= \sum_{i=1}^N \left(\frac{1}{1 + e^{-z_i}} - y_i \right) x_{1i} \\ \frac{\partial J}{\partial \beta_2} &= \sum_{i=1}^N \left(\frac{1}{1 + e^{-z_i}} - y_i \right) x_{2i}\end{aligned}$$

- (c) Can you find the close form expressions for the optimal parameters $\hat{\beta}_0$, $\hat{\beta}_1$, and $\hat{\beta}_2$ by putting the derivatives of $J(\boldsymbol{\beta})$ to 0? What methods can be used to optimize the loss function $J(\boldsymbol{\beta})$?

Solution: No, we cannot find closed-form expressions for the optimal parameters. Setting the derivatives to zero gives us:

$$\begin{aligned}\sum_{i=1}^N \left(\frac{1}{1 + e^{-z_i}} - y_i \right) &= 0 \\ \sum_{i=1}^N \left(\frac{1}{1 + e^{-z_i}} - y_i \right) x_{1i} &= 0 \\ \sum_{i=1}^N \left(\frac{1}{1 + e^{-z_i}} - y_i \right) x_{2i} &= 0\end{aligned}$$

These are transcendental equations (nonlinear in the parameters $\beta_0, \beta_1, \beta_2$ because z_i contains these parameters inside the exponential). There is no analytical solution to these equations.

However, the good news is that $J(\boldsymbol{\beta})$ is convex, so iterative methods will converge to the global minimum. Methods to optimize $J(\boldsymbol{\beta})$:

- Gradient descent
- Newton's method (or Newton-Raphson)
- Quasi-Newton methods (e.g., BFGS, L-BFGS)
- Coordinate descent
- Stochastic gradient descent